

# Digital Material Passport

ID **DMP-TEST-MULTIROW-001** - Version **1.0.0**  
Issue Date **2025-04-10** - Certificate Type **EN 10204 3.1**

| Manufacturer                                                                             | Customer                                                                                             | Business Transaction |                          |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------|--------------------------|
| Example Forge Works GmbH<br>Stahlweg 8<br>47051 Duisburg<br>DE<br>info@example-forge.com | Example Steel Trading GmbH<br>Musterstrasse 10<br>70173 Stuttgart<br>DE<br>info@example-customer.com | Order                | ORD-2025-3312 · 18000 kg |
|                                                                                          |                                                                                                      | Delivery             | DLV-2025-4455 · 18000 kg |

## Product

|                    |                    |                     |                                               |
|--------------------|--------------------|---------------------|-----------------------------------------------|
| Product Name       | Forged round bars  | Name (EN 10027-1)   | S355J2 +N                                     |
| Batch ID           | BATCH-2025-03-7832 | Number (EN 10027-2) | 1.0577 +N                                     |
| Surface Condition  | Rough turned       | Shape               | RoundBar - D 350 mm - Length 2800 mm          |
| Delivery Condition | +N - (Normalized)  | Marking             | Both ends stamped with grade and heat number. |
|                    |                    | Packaging           | Bundled, max 20 MT                            |
|                    |                    | Coloring            | blue                                          |
|                    |                    | Product Norm        | EN 10250-1                                    |
|                    |                    | Product Norm        | EN 10250-2                                    |

## Heat Treatment

Process: **Normalizing** - Lot: HT-LOT-2025-0315-B - Furnace: FURNACE-07 - Date: 2025-03-15

| Stage         | Temperature | Duration | Cooling | Atmosphere |
|---------------|-------------|----------|---------|------------|
| Austenitizing | 910 C       |          |         | Air        |

## Chemical Analysis

Heat Number **H-2025-7832** - Melting Process **EAF+LF+VD** - Casting Method **IngotCasting**

| Symbol | C    | Si   | Mn   | P     | S     | Cr   | Ni   | Mo   | Cu   | B      | Al    | V     | Sn    | Ti    | Sb     | Ti  | Sb   | N     | Ca     | Nb    |
|--------|------|------|------|-------|-------|------|------|------|------|--------|-------|-------|-------|-------|--------|-----|------|-------|--------|-------|
| Unit   | %    | %    | %    | %     | %     | %    | %    | %    | %    | %      | %     | %     | %     | %     | %      | ppm | %    | %     | %      | %     |
| Min    | 0.1  |      | 1.1  |       | 0.015 |      |      |      |      |        | 0.015 |       |       |       |        |     |      |       |        |       |
| Max    | 0.22 | 0.55 | 1.65 | 0.03  | 0.03  | 0.35 | 0.35 | 0.1  | 0.3  |        | 0.06  | 0.025 | 0.015 | 0.006 | 0.003  | 3.0 | 0.5  | 0.012 | 0.003  | 0.03  |
| Actual | 0.17 | 0.28 | 1.38 | 0.014 | 0.022 | 0.18 | 0.14 | 0.05 | 0.22 | 4.0E-4 | 0.028 | 0.012 | 0.006 | 0.004 | 0.0012 | 2.1 | 0.38 | 0.009 | 0.0018 | 0.015 |

| Symbol | Co   | W    | F1   |
|--------|------|------|------|
| Unit   | %    | %    | %    |
| Min    |      |      |      |
| Max    | 0.15 | 0.05 | 0.52 |
| Actual | 0.08 | 0.02 | 0.37 |

F1 = Cr+Ni+Mo: 0.37%

## Mechanical Properties

| Property                                                           | Symbol | Actual                                     | Minimum   | Maximum   | Method           |
|--------------------------------------------------------------------|--------|--------------------------------------------|-----------|-----------|------------------|
| Tensile Strength <sup>1</sup><br>Specimen: 1/2T, L - ID: TS-001    | Rm     | 548 N/mm²                                  | 440 N/mm² | 630 N/mm² | EN ISO 6892-1/B1 |
| Yield Strength <sup>1</sup><br>Specimen: 1/2T, L - ID: YS-001      | Re     | 372 N/mm²                                  | 320 N/mm² |           | EN ISO 6892-1    |
| Elongation <sup>1</sup><br>Specimen: 1/2T, L - ID: EL-L-001        | A      | 26 %                                       | 17 %      |           | EN ISO 6892-1/B1 |
| Elongation <sup>1</sup><br>Specimen: 1/2T, T - ID: EL-T-001        | A      | 20 %                                       | 14 %      |           | EN ISO 6892-1/B1 |
| Reduction of area <sup>1</sup><br>Specimen: 1/2T, L - ID: RA-001   | Z      | 48 %                                       |           |           | EN ISO 6892-1    |
| Impact test ISO-V <sup>2</sup><br>Specimen: 1/2T, T - ID: IT-T-001 |        | 36 / 30 / 37 J - Average<br>34.33 - Min 22 | 22 J      |           | EN ISO 148-1     |
| Impact test ISO-V <sup>2</sup><br>Specimen: 1/2T, L - ID: IT-L-001 |        | 62 / 65 / 58 J - Average<br>61.67 - Min 38 | 38 J      |           | EN ISO 148-1     |
| Surface hardness                                                   |        | 162 HB                                     |           | 180 HB    |                  |

<sup>1</sup> in delivery condition <sup>2</sup> -20°C, in delivery condition

Physical Properties

| Property           | Actual   | Target/Min | Maximum  | Method    |
|--------------------|----------|------------|----------|-----------|
| Grain size         | 6        | 4          | 9        | ASTM E112 |
| Purity K4(ox)      | 20       |            | 35       | DIN 50602 |
| Purity K4(s)       | 25       |            | 35       | DIN 50602 |
| Straightness       | 1.2 mm/m |            | 2.0 mm/m |           |
| Reduction ratio    | 3.8      | 3.0        |          |           |
| Residual magnetism | 5 A/cm   |            | 10 A/cm  |           |

✓ fully formed structure, no residue of dendritic cast structure

Supplementary Tests

| Test                    | Result / limits     | Method             |
|-------------------------|---------------------|--------------------|
| Ultrasonic test<br>100% | Yes - Pass          | EN 10228-3 class 3 |
| Radioactivity           | Yes - max 100 Bq/kg |                    |

✓ 100% visually inspected, surface free from defects

Validation

We hereby certify that the material described above has been manufactured, tested and inspected in accordance with the requirements of the applicable specifications and standards, and that the results comply with the specified requirements.

Validated by **Dr. Maria Fischer**, Head of Quality Assurance, Quality Assurance - 2025-04-10

✓ Material is of non-Russian origin (EU Regulation No. 833/2014)